

Rational points and inflexions on Hermitian-relative curves

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Abstract

In [2], we introduced the notion of a Hermitian-relative curve, which is a plane curve defined by $(x^{\sqrt{q}}, y^{\sqrt{q}}, z^{\sqrt{q}})A(x, y, z)^t = 0$ with $A \in GL(3, \mathbb{F}_q)$. After investigating their basic properties, we classified those curves with two or more rational inflexions. In this paper, we first complete the classification of curves whose rational points are all inflexions by demonstrating the existence of curves with exactly one rational point. As a continuation, we then classify Hermitian-relative curves that possess at least one rational point which is not an inflexion. We categorize these curves according to the ordered pair consisting of the number of rational points and the number of inflexions. Finally, we enumerate all possible pairs and prove that each pair is realized by a suitable curve.

Key Words : Plane curve, Finite field, Hermitian-relative curve, Rational point, Inflexion.

1 Introduction

Let q be an even power of a prime p . In [2], we introduced *relatives of the Hermitian curve* (briefly, *Hermitian-relative curves*) and established several fundamental properties of this family of curves. In [3], Pardini proved that these curves are equivalent to the Hermitian curve over the algebraic closure

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$\overline{\mathbb{F}}_q$ of $\mathbb{F}_q$. In [2], we showed that every such curve has at least one $\mathbb{F}_q$ -rational point, and moreover that the number of these points is congruent to 1 modulo $\sqrt{q}$. In particular, we classified those curves having two or more rational inflexions.

It is well known that the Hermitian curve has $q\sqrt{q} + 1$ rational points, all of which are inflexions. In addition to this classical case, we previously classified curves with $\sqrt{q} + 1$ rational points, all being inflexions, in [2]. To complete the classification of curves whose rational points are all inflexions, we characterize the remaining case—curves with exactly one rational point—in the following theorem:

Theorem 1.1. *A Hermitian-relative curve C has exactly one rational point if and only if it is projectively equivalent to the curve C_η defined by the equation $x^{\sqrt{q}+1} + \eta y^{\sqrt{q}+1} + y^{\sqrt{q}}z + yz^{\sqrt{q}} = 0$ for some $\eta \in \mathbb{F}_q \setminus \mathbb{F}_{\sqrt{q}}$. In this case, the unique rational point is an inflexion.*

For a curve C , we denote $N_q(C)$ and $I_q(C)$ the numbers of rational points and rational inflexions on C , respectively. We say that a curve C is of type (n, i) if $N_q(C) = n$ and $I_q(C) = i$. The above curves are of type $(q\sqrt{q} + 1, q\sqrt{q} + 1)$, $(\sqrt{q} + 1, \sqrt{q} + 1)$, and $(1, 1)$, respectively.

Theorem 1.2. *There exist Hermitian-relative curves of types $(q\sqrt{q}+1, q\sqrt{q}+1)$, $(\sqrt{q} + 1, \sqrt{q} + 1)$, and $(1, 1)$. Moreover, if a Hermitian-relative curve C satisfies $N_q(C) = I_q(C)$, then its type must be one of these three cases.*

Now we consider the curves of type (n, i) with $n \neq i$. We classify the curves that possess at least one rational point which is not an inflexion. We show that all such curves are equivalent to curves of the form

$$C_{a,b} : x^{\sqrt{q}}z + xy^{\sqrt{q}} + bxz^{\sqrt{q}} + ayz^{\sqrt{q}} = 0,$$

where $a, b \in \mathbb{F}_q$, $a \neq 0$.

We first classify the curves with $b = 0$, i.e., the curves

$$C_{a,0} : x^{\sqrt{q}}z + xy^{\sqrt{q}} + ayz^{\sqrt{q}} = 0,$$

where $a \in \mathbb{F}_q$, $a \neq 0$.

Theorem 1.3. *For the curve $C_{a,0}$, the following statements hold:*

- (1) *If $\sqrt{q} \equiv 0 \pmod{3}$, then $C_{a,0}$ is of type $(q + 1, 1)$.*
- (2) *If $\sqrt{q} \equiv 1 \pmod{3}$, then $C_{a,0}$ is of type $(q + 1, 2)$.*
- (3) *If $\sqrt{q} \equiv 2 \pmod{3}$, then $C_{a,0}$ is of type $(q + 2\sqrt{q} + 1, 0)$ for $a = \omega^{3j}$, $0 \leq j < \frac{q-1}{3}$ and $(q - \sqrt{q} + 1, 0)$, otherwise. Here ω is a primitive element of $\mathbb{F}_q$.*

When $b \neq 0$, the curve $C_{a,b}$ is equivalent to a curve of the form C_c with a single parameter $c \in \mathbb{F}_q^*$ defined by the equation:

$$C_c : x^{\sqrt{q}}z - cxy^{\sqrt{q}} + cxz^{\sqrt{q}} - yz^{\sqrt{q}} = 0$$

which corresponds to the matrix $M_c = \begin{pmatrix} 0 & 0 & 1 \\ -c & 0 & 0 \\ c & -1 & 0 \end{pmatrix}$.

Since $c \in \mathbb{F}_q^*$ it suffices to study the $q-1$ matrices and their corresponding $q-1$ curves.

To investigate that in the next section, we introduce some definitions and notations. Recall that for a matrix $A = (a_{ij})$, $A^{(\sqrt{q})}$ denote the matrix $(a_{ij}^{\sqrt{q}})$ and $A^* = (A^{(\sqrt{q})})^t$ is the transpose of $A^{(\sqrt{q})}$.

For a Hermitian-relative curve C_A corresponding to A , the curve C_{A^*} is called the *mirror curve* of C_A . Of course, the mirror curve of C_{A^*} is the original one C_A , and C_A is Hermitian if and only if $C_{A^*} = C_A$.

Lemma 1.4. *For a mirror pair (C_A, C_{A^*}) , the following properties hold. ([2, Prop. 2.6])*

- (1) $C_A(\mathbb{F}_q) = C_{A^*}(\mathbb{F}_q)$.
- (2) For $P \in C_A(\mathbb{F}_q) = C_{A^*}(\mathbb{F}_q)$, P is an inflexion of C_A if and only if P is an inflexion of C_{A^*} . In this case, $\mathcal{L}_P C_A = \mathcal{L}_P C_{A^*}$, where $\mathcal{L}_P C$ is the tangent line to C at P .
- (3) For a non-inflexion $P \in C_A(\mathbb{F}_q)$, there is a point $P' \in C_A(\mathbb{F}_q)$ so that $C_A \cdot \mathcal{L}_P C_A = \sqrt{q}P + P'$. In this case, $C_{A^*} \cdot \mathcal{L}_{P'} C_{A^*} = \sqrt{q}P' + P$. In particular P' is a non-inflexion of C_{A^*} .

Recall that among nonsingular plane curves of degree $\sqrt{q}+1$ with $q > 4$, a Hermitian-relative curve C is characterized by the following property: for any point $P \in C(\overline{\mathbb{F}}_q)$, the intersection multiplicity $i(C \cdot \mathcal{L}_P C; P)$ of C and its tangent line $\mathcal{L}_P C$ at P is either $\sqrt{q}$ or $\sqrt{q}+1$ [1].

Lemma 1.5. *Let C_A be a Hermitian-relative curve which is not a Hermitian curve. Then the following statements hold.*

- (1) $\sum_{P \in C_A(\mathbb{F}_q)} i(C_A \cdot C_{A^*}; P) \leq \sum_{P \in C_A(\overline{\mathbb{F}}_q)} i(C_A \cdot C_{A^*}; P) = (\sqrt{q}+1)^2$.
- (2) If P is an inflexion, then $i(C_A \cdot C_{A^*}; P) \geq \sqrt{q}+1$.
- (3) $N_q(C_A) \leq (\sqrt{q}+1)^2$.

We may consider a map $\varphi_A : C_A(\mathbb{F}_q) \rightarrow C_A(\mathbb{F}_q)$ by $\varphi(P) = P'$, where $C_A \cdot \mathcal{L}_P C_A = \sqrt{q}P + P'$. Note that if P is an inflexion, then $\varphi(P) = P$. By simple computation, we have $\varphi(P) = (A^{-1}A^*)P$ as a point in the projective plane. Hence any inflexion of C_A is an eigenvector of the matrix $A^{-1}A^*$.

2 Main Results

For $c \in \mathbb{F}_q^*$, we consider a curve C_c defined by the equation

$$C_c : x^{\sqrt{q}}z - cxy^{\sqrt{q}} + cxz^{\sqrt{q}} - yz^{\sqrt{q}} = 0,$$

which corresponds to the matrix $M_c = \begin{pmatrix} 0 & 0 & 1 \\ -c & 0 & 0 \\ c & -1 & 0 \end{pmatrix}$. Let $U_c = M_c^{-1}M_c^* =$

$$\begin{pmatrix} 0 & 0 & \frac{1}{c} \\ -1 & 0 & 1 \\ 0 & -c\sqrt{q} & c\sqrt{q} \end{pmatrix}. \text{ Then the characteristic polynomial of } U_c \text{ is}$$

$$P_{U_c}(\lambda) = \lambda^3 - c\sqrt{q}\lambda^2 + c\sqrt{q}\lambda - c\sqrt{q}^{-1}.$$

The curve C_c meets the line $z = 0$ at two points $(1, 0, 0)^t$ and $(0, 1, 0)^t$. From now on, we consider only the affine part C_c° of the curve C_c given by

$$C_c^\circ : x^{\sqrt{q}} - cxy^{\sqrt{q}} + cx - y = 0.$$

Lemma 2.1. *For an eigenvalue λ_0 of U_c , $(\frac{1}{c\lambda_0}, 1 - c^{-\sqrt{q}}\lambda_0, 1)^t = (\frac{1}{c\lambda_0}, \frac{-1+c\lambda_0}{c\lambda_0^2}, 1)^t$ is the corresponding eigenvector.*

Lemma 2.2. *If $(x_0, y_0, 1)^t$ is an inflexion, then $\frac{1}{cx_0}$ is a root of $P_{U_c}(\lambda)$. If the line $y = y_0$ is a $\sqrt{q}$ -line of the affine part C_c° of C_c , then $c\sqrt{q}(1 - y_0)$ is a root of $P_{U_c}(\lambda)$.*

If an eigenvector of U_c is a point on C_c , then it is an inflexion of C_c . The number of rational points on C_c° is expressed as $q - \sqrt{q} - 1 + s\sqrt{q}$, and hence $N_q(C_c) = q - \sqrt{q} + 1 + s\sqrt{q}$, where s is the number of $\sqrt{q}$ -lines of C_c° with the equation $y = y_0$. So it suffices to find the roots of $P_{U_c}(\lambda)$.

Recall that two matrices A and B are said to be similar if there exists a nonsingular matrix T such that $B = T^{-1}AT$. It is also a well-known fact that similar matrices share the same characteristic polynomial.

Lemma 2.3. *Two matrices U_c^* and U_c^{-1} are similar to each other. So we have $P_{U_c^*}(\lambda) = P_{U_c^{-1}}(\lambda)$. Therefore, if λ_0 is an eigenvalue of U_c , then $\lambda_0^{\sqrt{q}} = \lambda_1^{-1}$ for some eigenvalue λ_1 of U_c . Note that λ_0 need not be equal to λ_1 .*

Theorem 2.4. *For each $c \in \mathbb{F}_q^*$, the polynomial $P_{U_c}(\lambda)$ either splits completely into linear factors or is irreducible over $\mathbb{F}_q$. More precisely, one of the following statements holds:*

- (i) $P_{U_c}(\lambda)$ has the triple root $\lambda = 1$. This occurs only when $c = 3$ and $p \neq 3$ since we assumed $c \neq 0$.

- (ii) $P_{U_c}(\lambda)$ has one double root and the other simple root, both of which are $(\sqrt{q} + 1)$ -th roots of unity.
- (iii) $P_{U_c}(\lambda)$ has three simple roots, all of which are $(\sqrt{q} + 1)$ -th roots of unity.
- (iv) $P_{U_c}(\lambda)$ has three simple roots, only one of which is a $(\sqrt{q} + 1)$ -th root of unity, and the other two, say β and γ , satisfy $\beta\sqrt{q} = \gamma^{-1}$.
- (v) $P_{U_c}(\lambda)$ is irreducible over $\mathbb{F}_q$.

Notation. Let μ_{q+1} denote the set of $(\sqrt{q} + 1)$ -th roots of unity in $\mathbb{F}_q$, i.e., $\mu_{\sqrt{q}+1} = \{\zeta \in \mathbb{F}_q \mid \zeta^{\sqrt{q}+1} = 1\}$.

Lemma 2.5. *If $\alpha \in \mathbb{F}_q^*$ is a multiple root of $P_{U_c}(\lambda)$, then $\alpha \in \mu_{\sqrt{q}+1}$ and the eigenvector corresponding to it represents a point on the curve C_c , which is an inflexion.*

Lemma 2.6. *If α is a root of $P_{U_c}(\lambda)$ and $\alpha \in \mu_{\sqrt{q}+1}$, then the corresponding line $y = 1 - c^{-\sqrt{q}}\alpha$ is a $\sqrt{q}$ -line of C_c° .*

Lemma 2.7. *If $P_{U_c}(\lambda)$ has three distinct roots α_i , $i = 1, 2, 3$, which belong to $\mu_{\sqrt{q}+1}$, then the three corresponding lines $y = 1 - c^{-\sqrt{q}}\alpha_i$ are all $\sqrt{q}$ -lines of C_c° and C_c has no inflexion.*

Lemma 2.8. *Suppose that $P_{U_c}(\lambda)$ has three distinct roots, exactly one of which belongs to $\mu_{\sqrt{q}+1}$. Then the two eigenvectors corresponding to the other two eigenvalues are inflexions of C_c .*

Up to this point, we have used $P_{U_c}(\lambda)$ to denote the characteristic polynomial; however, hereafter we shall adopt the notation $f_c(\lambda)$ to emphasize our focus on the parameter c .

Lemma 2.9. *Let $\alpha, \beta, \gamma \in \mu_{\sqrt{q}+1}$. If $\alpha + \beta + \gamma = c^{\sqrt{q}} = \alpha\beta + \beta\gamma + \alpha\gamma$ for some $c \in \mathbb{F}_q^*$, then it necessarily follows that $\alpha\beta\gamma = c^{\sqrt{q}-1}$. Consequently, such α, β , and γ are precisely the roots of the polynomial f_c .*

Lemma 2.10. *Let $p \neq 2, 3$ and $\alpha \in \mu_{\sqrt{q}+1}$. Then there exists an element $c \in \mathbb{F}_q^*$ such that α is a double root of the polynomial f_c . Moreover, the remaining root of f_c is given by $\frac{\alpha(2-\alpha)}{2\alpha-1}$, which also belongs to $\mu_{\sqrt{q}+1}$.*

Lemma 2.11. *For the cases $p = 2$ and $p = 3$, the following statements hold.*

- (1) *When $p = 2$, for each $\alpha \in \mu_{\sqrt{q}+1}$, the polynomial f_c , $c = \alpha^{2\sqrt{q}}$, has α as a double root and the other root is α^2 . Note that if $\alpha = 1$, it is a triple root.*

- (2) When $p = 3$, if the polynomial f_c (with $c \neq 0$) possesses a double root α , then it must be $\alpha = 2$. Specifically, for $\alpha = 2 \in \mu_{\sqrt{q}+1}$ and any $\beta \in \mu_{\sqrt{q}+1} \setminus \{2\}$, the polynomial f_c with $c = (1 + \beta)\sqrt{q}$ has 2 as a double root and β as the remaining simple root. Note that the case $\beta = 2$ is excluded because it leads to $c = 0$, which contradicts our assumption that $c \in \mathbb{F}_q^*$.

Remark 2.12. The polynomial f_c has a triple root if and only if $\alpha = \frac{\alpha(2-\alpha)}{2\alpha-1}$, which is equivalent to $\alpha = 1$ since $\alpha \neq 0$. In fact, when $c = 3$, we have $f_c(\lambda) = \lambda^3 - c\sqrt{q}\lambda^2 + c\sqrt{q}\lambda - c\sqrt{q-1} = \lambda^3 - 3\lambda^2 + 3\lambda - 1 = (\lambda - 1)^3$.

Lemma 2.13. Let α and $\beta \in \mu_{\sqrt{q}+1}$. Suppose that α and β are roots of f_c for some $c \in \mathbb{F}_q^*$.

- (1) If $\alpha + \beta \neq 1$ and $\alpha^2 + \alpha\beta + \beta^2 \neq 0$, then the third root γ also belongs to $\mu_{\sqrt{q}+1}$. In this case, γ and c are explicitly given by

$$\gamma = \frac{\alpha + \beta - \alpha\beta}{\alpha + \beta - 1} \quad \text{and} \quad c = \left(\frac{\alpha^2 + \alpha\beta + \beta^2}{\alpha + \beta - 1} \right)^{\sqrt{q}}.$$

- (2) If $\alpha + \beta = 1$, then it necessarily follows that $\alpha\beta = 1$. Furthermore, for any $\gamma \in \mu_{\sqrt{q}+1} \setminus \{-1\}$, the elements α, β, γ are the roots of $f_c(\lambda)$ with $c = (1 + \gamma)\sqrt{q} = 1 + \gamma\sqrt{q}$.

As before, ω denotes a primitive element of $\mathbb{F}_q$.

Lemma 2.14. The solutions to the system of equations $\begin{cases} \alpha\beta = 1 \\ \alpha + \beta = 1 \end{cases}$ over $\mathbb{F}_q$ are given as follows:

- (1) If q is even (i.e., $p = 2$), then $\{\alpha, \beta\} = \left\{ \omega^{\frac{q-1}{3}}, \omega^{\frac{2(q-1)}{3}} \right\}$.

- (2) If $3 \mid q$ (i.e., $p = 3$), then $\alpha = \beta = -1 = 2$.

- (3) If $p \neq 2, 3$, then $\{\alpha, \beta\} = \left\{ \omega^{\frac{q-1}{6}}, \omega^{\frac{5(q-1)}{6}} \right\}$.

Lemma 2.15. Let α be a $(\sqrt{q} + 1)$ -th root of unity (i.e., $\alpha^{\sqrt{q}+1} = 1$). Let β and γ be elements of $\mathbb{F}_q$ satisfying $\gamma = \beta^{-\sqrt{q}}$. If $\alpha + \beta + \gamma = c\sqrt{q} = \alpha\beta + \beta\gamma + \gamma\alpha$ for some $c \in \mathbb{F}_q$, then $\alpha\beta\gamma = c\sqrt{q-1}$. In particular, α, β, γ are the roots of the polynomial $f_c(\lambda)$.

Lemma 2.16. Let β and γ be elements of $\mathbb{F}_q$ such that $\beta\sqrt{q} = \gamma^{-1}$ (or equivalently, $\beta\gamma\sqrt{q} = 1$).

- (1) If $\beta + \gamma \neq 1$ and $\beta^2 + \beta\gamma + \gamma^2 \neq 0$, let $\alpha = \frac{\beta + \gamma - \beta\gamma}{\beta + \gamma - 1}$. Then α, β, γ are the roots of the polynomial f_c for $c = \left(\frac{\beta^2 + \beta\gamma + \gamma^2}{\beta + \gamma - 1} \right)^{\sqrt{q}}$. Moreover, α is a $(\sqrt{q} + 1)$ -th root of unity.

- (2) If $\beta + \gamma = 1$, then $\gamma = \beta^{-1} \in \mathbb{F}_{\sqrt{q}}$ and the roots of the polynomial f_c are $\{c^{\sqrt{q}-1}, \beta, \gamma\}$, where c satisfies the relation $c^{\sqrt{q}} - c^{\sqrt{q}-1} - 1 = 0$. Indeed, this equation has exactly $\sqrt{q}$ distinct roots for c .

Lemma 2.17. Let α and β be $(\sqrt{q} + 1)$ -th roots of unity in $\mathbb{F}_q$. Suppose that $\alpha + \beta = 1$. Then $\alpha\beta = 1$ and the following statements hold:

- (1) The case $q \equiv 0 \pmod{2}$ (Characteristic $p = 2$): If $\sqrt{q} \equiv 2 \pmod{3}$, then the solutions are:

$$\begin{cases} \alpha = \omega^{\frac{q-1}{3}} \\ \beta = \alpha^{-1} = \omega^{\frac{2(q-1)}{3}} \end{cases} \quad \text{or} \quad \begin{cases} \alpha = \omega^{\frac{2(q-1)}{3}} \\ \beta = \alpha^{-1} = \omega^{\frac{q-1}{3}} \end{cases}$$

If $\sqrt{q} \equiv 1 \pmod{3}$, then there is no solution.

- (2) The case $q \equiv 0 \pmod{3}$ (Characteristic $p = 3$): The only solution is $\alpha = \beta = 2$.

- (3) The other cases ($p \neq 2, 3$): If $\sqrt{q} \equiv 2 \pmod{3}$, then the solutions are:

$$\begin{cases} \alpha = \omega^{\frac{q-1}{6}} \\ \beta = \alpha^{-1} = \omega^{\frac{5(q-1)}{6}} \end{cases} \quad \text{or} \quad \begin{cases} \alpha = \omega^{\frac{5(q-1)}{6}} \\ \beta = \alpha^{-1} = \omega^{\frac{q-1}{6}} \end{cases}$$

If $\sqrt{q} \equiv 1 \pmod{3}$, then there is no solution.

Lemma 2.18. Let α and β be elements of $\mathbb{F}_q$ such that $\alpha^{\sqrt{q}} = \beta^{-1}$ and $\alpha + \beta = 1$. Then the following statements hold:

- (1) Case $q \equiv 0 \pmod{2}$: If $\sqrt{q} \equiv 1 \pmod{3}$, the solutions are:

$$\begin{cases} \alpha = \omega^{\frac{q-1}{3}} \\ \beta = \alpha^{-1} = \omega^{\frac{2(q-1)}{3}} \end{cases} \quad \text{or} \quad \begin{cases} \alpha = \omega^{\frac{2(q-1)}{3}} \\ \beta = \alpha^{-1} = \omega^{\frac{q-1}{3}} \end{cases}$$

If $\sqrt{q} \equiv 2 \pmod{3}$, there is no solution.

- (2) Case $q \equiv 0 \pmod{3}$: The unique solution is $\alpha = \beta = 2$, which belongs to the group of $(\sqrt{q} + 1)$ -th roots of unity, $\mu_{\sqrt{q}+1}$.

- (3) Case $p \neq 2, 3$: If $\sqrt{q} \equiv 1 \pmod{3}$, the solutions are:

$$\begin{cases} \alpha = \omega^{\frac{q-1}{6}} \\ \beta = \alpha^{-1} = \omega^{\frac{5(q-1)}{6}} \end{cases} \quad \text{or} \quad \begin{cases} \alpha = \omega^{\frac{5(q-1)}{6}} \\ \beta = \alpha^{-1} = \omega^{\frac{q-1}{6}} \end{cases}$$

If $\sqrt{q} \equiv 2 \pmod{3}$, there is no solution.

Lemma 2.19. Consider the equation $\alpha^2 + \alpha\beta + \beta^2 = 0$ for $\alpha, \beta \in \mu_{\sqrt{q}+1}$.

- (1) Case $\sqrt{q} \equiv 0 \pmod{3}$: The equation is satisfied if and only if $\alpha = \beta$.
- (2) Case $\sqrt{q} \equiv 1 \pmod{3}$: The equation has no solutions in $\mu_{\sqrt{q}+1}$.
- (3) Case $\sqrt{q} \equiv 2 \pmod{3}$: The solutions are $\beta = \alpha\omega^{\frac{q-1}{3}}$ and $\beta = \alpha\omega^{\frac{2(q-1)}{3}}$.

Lemma 2.20. Consider the equation $\alpha^2 + \alpha\beta + \beta^2 = 0$ for $\alpha, \beta \in \mathbb{F}_q$ such that $\alpha\sqrt{q} = \beta^{-1}$. Then the following statements hold:

- (1) Case $\sqrt{q} \equiv 0 \pmod{3}$: The equation is satisfied if and only if $\alpha = \beta$. In this case, α (and thus β) necessarily belongs to $\mu_{\sqrt{q}+1}$.
- (2) Case $\sqrt{q} \equiv 1 \pmod{3}$: The solutions are given by $\beta = \alpha\omega^{\frac{q-1}{3}}$ or $\beta = \alpha\omega^{\frac{2(q-1)}{3}}$.
- (3) Case $\sqrt{q} \equiv 2 \pmod{3}$: There are no solutions satisfying the given conditions.

Lemma 2.21. Let three sets as follows;

$$S_1 = \{(\alpha, \beta, \gamma) \in (\mu_{\sqrt{q}+1})^3 \mid \alpha, \beta, \gamma \text{ are roots of } f_c \text{ for some } c \in \mathbb{F}_q^*\}$$

$$S_2 = \{(\alpha, \beta, \gamma) \in (\mu_{\sqrt{q}+1})^3 \mid \alpha + \beta \neq 1, \alpha^2 + \alpha\beta + \beta^2 \neq 0, \gamma = \frac{\alpha + \beta - \alpha\beta}{\alpha + \beta - 1}\}$$

$$S_3 = \{(\alpha, \beta, \gamma) \in (\mu_{\sqrt{q}+1})^3 \mid \alpha + \beta = 1, \gamma \neq -1\}.$$

Then S_1 is the disjoint union of S_2 and S_3 .

Lemma 2.22. Assume that $\sqrt{q} \equiv 0 \pmod{3}$. Under this condition, we observe the following properties regarding the polynomials f_c :

- (1) There exists no polynomial f_c with $c \neq 0$ that possesses a triple root.
- (2) The number of polynomials f_c having a double root is $\sqrt{q}$. In this case, both the double root and the remaining simple root belong to $\mu_{\sqrt{q}+1}$.
- (3) The number of polynomials f_c that have three distinct roots, all of which belong to $\mu_{\sqrt{q}+1}$, is given by $\frac{q - \sqrt{q}}{6}$.

Lemma 2.23. Assume that $\sqrt{q} \equiv 1 \pmod{3}$. Under this condition, we observe the following properties regarding the characteristic polynomials f_c :

- (1) There exists a unique polynomial f_c with $c \neq 0$ that has a triple root. In this case, the triple root is 1 and $c = 3$ (or $c = 1$ if $p = 2$).
- (2) The number of polynomials f_c having a double root is $\sqrt{q}$. In this case, both the double root and the remaining simple root belong to $\mu_{\sqrt{q}+1}$.

- (3) The number of polynomials f_c that have three distinct roots, all of which belong to $\mu_{\sqrt{q}+1}$, is given by $\frac{q - \sqrt{q}}{6}$.

Lemma 2.24. Assume that $\sqrt{q} \equiv 2 \pmod{3}$. Under this condition, we observe the following properties regarding the polynomials f_c :

- (1) There exists a unique polynomial f_c with $c \neq 0$ that has a triple root. In this case, the triple root is 1 and $c = 3$ (or $c = 1$ if $p = 2$).
- (2) The number of polynomials f_c having a double root is $\sqrt{q}$. In this case, both the double root and the remaining simple root belong to $\mu_{\sqrt{q}+1}$.
- (3) The number of polynomials f_c that have three distinct roots, all of which belong to $\mu_{\sqrt{q}+1}$, is given by $\frac{q - \sqrt{q} - 2}{6}$.

Lemma 2.25. The number of characteristic polynomials that have three distinct simple roots, where only one of them belongs to $\mu_{\sqrt{q}+1}$, is:

$$\begin{cases} \frac{q - \sqrt{q} - 2}{2} & \text{when } \sqrt{q} \equiv 0 \pmod{3} \\ \frac{q - \sqrt{q} - 4}{2} & \text{when } \sqrt{q} \equiv 1 \pmod{3} \\ \frac{q - \sqrt{q} - 2}{2} & \text{when } \sqrt{q} \equiv 2 \pmod{3} \end{cases}$$

Lemma 2.26. The number of irreducible characteristic polynomials f_c , $c \in \mathbb{F}_q^*$ over $\mathbb{F}_q$ is:

$$\begin{cases} \frac{q - \sqrt{q}}{3} & \text{when } \sqrt{q} \equiv 0 \pmod{3} \\ \frac{q - \sqrt{q}}{3} & \text{when } \sqrt{q} \equiv 1 \pmod{3} \\ \frac{q - \sqrt{q} - 2}{3} & \text{when } \sqrt{q} \equiv 2 \pmod{3} \end{cases}$$

Theorem 2.27. Using abover lemmas, we obtain the following table:

In the table, “Cases” are defined as follows;

- (I) f_c is an irreducible polynomial.
- (II) f_c has a triple root.
- (III) f_c has three simple roots, only one of them is $(\sqrt{q}+1)$ -th root of unity.
- (IV) f_c has a double root and the other simple root, both of them are $(\sqrt{q}+1)$ -th roots of unity.
- (V) f_c has three simple roots which are all $(\sqrt{q}+1)$ -th roots of unity.

Case	type (n, i)	the number of polynomials		
		$\sqrt{q} \equiv 0 \pmod{3}$	$\sqrt{q} \equiv 1 \pmod{3}$	$\sqrt{q} \equiv 2 \pmod{3}$
(I)	$(q - \sqrt{q} + 1, 0)$	$\frac{q - \sqrt{q}}{3}$	$\frac{q - \sqrt{q}}{3}$	$\frac{q - \sqrt{q} - 2}{3}$
(II)	$(q + 1, 1)$	0	1	1
(III)	$(q + 1, 2)$	$\frac{q - \sqrt{q} - 2}{2}$	$\frac{q - \sqrt{q} - 4}{2}$	$\frac{q - \sqrt{q} - 2}{2}$
(IV)	$(q + \sqrt{q} + 1, 1)$	$\sqrt{q}$	$\sqrt{q}$	$\sqrt{q}$
(V)	$(q + 2\sqrt{q} + 1, 0)$	$\frac{q - \sqrt{q}}{6}$	$\frac{q - \sqrt{q}}{6}$	$\frac{q - \sqrt{q} - 2}{6}$

Theorem 2.28. *For any q , there exist curves of exactly eight types, $(q\sqrt{q} + 1, q\sqrt{q} + 1)$, $(\sqrt{q} + 1, \sqrt{q} + 1)$, $(1, 1)$, $(q - \sqrt{q} + 1, 0)$, $(q + 1, 1)$, $(q + 1, 2)$, $(q + \sqrt{q} + 1, 1)$, and $(q + 2\sqrt{q} + 1, 0)$.*

proof) From Theorems 1.2, 1.3, and 2.27, we obtain the result.

References

- [1] M. Homma, *Funny plane curves in characteristic $p > 0$* , Comm. Algebra 15 (1987) 1469–1501.
- [2] M. Homma and S.J. Kim, *Relatives of the Hermitian curve*, Journal of Geometry(2026) 117:11. <https://doi.org/10.1007/s00022-026-00795-8>.
- [3] R. Pardini, *Some remarks on plane curves over fields of finite characteristic*, Compositio Math. 60 (1986) 3–17.